

## Week 4 Module:

Date: 09/07/2026

### EASY LEVEL:

#### Problem Statement:

Given a positive integer **N**, determine the total number of digits present in the number.

Your task is to write a program that counts and prints the total number of digits.

#### Input Format

A single integer **N**.

#### Output Format

Print a single integer representing the total number of digits in **N**.

#### Constraints

$$1 \leq N \leq 10^{18}$$

#### Example 1

##### Input

12345

##### Output

5

##### Explanation

The number **12345** contains **5** digits.

#### Example 2

##### Input

7

##### Output

1

##### Explanation

The number **7** consists of only one digit.

```
#include <stdio.h>
#include <conio.h>
```

```
void main()
```

```
{
```

```
    long long int n;
    int count = 0;
```

```
    clrscr();
```

```
    printf("enter a number:");
    scanf("%lld", &n);
```

```
    while (n > 0)
```

```
    {
```

```
        count ++;
```

```
        n = n / 10;
```

```
    }
```

```
    printf("Total digits = %d",
           count);
```

getch (r);

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ADHAYA.CPP

```
#include<stdio.h>
#include<conio.h>
```

```
void main()
{
    long long int n;
    int count=0;

    clrscr();

    printf("enter a number:");
    scanf("%lld", &n);

    while (n>0)
    {
        count++;
        n=n/10;
    }
    printf("total digits = %d", count);
    getch();
}
```

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```
enter a number:12345  
total digits = 5_
```

**Problem Statement**

Given a positive integer **N**, reverse its digits and print the reversed number.

Your task is to write a program that reverses the given number.

**Input Format**

A single integer **N**.

**Output Format**

Print the reversed number.

**Constraints**

$1 \leq N \leq 10^{18}$

**Example 1****Input**

12345

**Output**

54321

**Explanation**

Reversing the digits of **12345** gives **54321**.

**Example 2****Input**

90010

**Output**

1009

**Explanation**

After reversing the digits, the leading zeros are removed automatically, resulting in **1009**.

```
#include<stdio.h>
#include<conio.h>
```

```
void main()
{
    long long int n,rev=0,rem;

    clrscr();

    printf("enter a number:");
    scanf("%lld", &n);

    while (n>0)
    {
        rem=n%10;
        rev=rev*10+rem;
        n=n/10;
    }
    printf("reversed number=%lld",rev);
    getch();
}
```



```
enter a number:12345  
reversed number=54321_
```



**Problem Statement**

Given a positive integer **N**, calculate the sum of all its digits.

Your task is to write a program that computes and prints the sum of the digits.

**Input Format**

A single integer **N**.

**Output Format**

Print a single integer representing the sum of all digits.

**Constraints**

$$1 \leq N \leq 10^{18}$$

**Example 1****Input**

12345

**Output**

15

**Explanation**

The sum of the digits is:

$$1 + 2 + 3 + 4 + 5 = 15$$

**Example 2****Input**

9087

**Output**

24

**Explanation**

The sum of the digits is:

$$9 + 0 + 8 + 7 = 24$$

```
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ADHAYA.CPP

#include<stdio.h>
#include<conio.h>

void main()
{
    long long int n;
    int sum=0,rem;

    clrscr();

    printf("enter a number:");
    scanf("%lld", &n);

    while (n>0)
    {
        rem=n%10;
        sum=sum+rem;
        n=n/10;
    }
    printf("sum of digits=%d",sum);
    getch();
}

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```

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```
    getch();
```

```
}
```

enter a number:12345

sum of digits=15



## Problem Statement

Given a positive integer **N**, count how many **even** digits and how many **odd** digits are present in the number.

Your task is to write a program that prints the total count of even digits followed by the total count of odd digits.

## Input Format

A single integer **N**.

## Output Format

Print two integers separated by a space:

- Number of even digits
- Number of odd digits

## Constraints

$$1 \leq N \leq 10^{18}$$

## Example 1

### Input

123456

### Output

3 3

### Explanation

Even digits are **2, 4, 6**.

Odd digits are **1, 3, 5**.

Therefore, the output is **3 3**.

## Example 2

### Input

908172

### Output

3 3

### Explanation

Even digits are **0, 8, 2**.

Odd digits are **9, 1, 7**.

Therefore, the output is **3 3**.

---



```
#include<stdio.h>
#include<conio.h>
```

```
void main()
{
    long long int n;
    int digit,even=0,odd=0;

    clrscr();

    printf("enter a number:");
    scanf("%lld", &n);

    while (n>0)
    {
        digit=n%10;

        if(digit%2==0)
            even++;
        else
            odd++;
    }
}
```

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```
    odd++;
```

```
    n=n/10;
```

```
}
```

```
printf("even digits=%d\n",even);
```

```
printf("odd digits=%d",odd);
```

```
    getch();
```

```
}
```

enter a number:123456

even digits=3

odd digits=3\_



**Problem Statement**

Given a positive integer **N**, find and print the largest digit present in the number.

Your task is to write a program that determines the maximum digit.

**Input Format**

A single integer **N**.

**Output Format**

Print the largest digit present in the given number.

**Constraints**

$$1 \leq N \leq 10^{18}$$

**Example 1****Input**

472953

**Output**

9

**Explanation**

The digits are **4, 7, 2, 9, 5, 3**.

The largest digit among them is **9**.

**Example 2****Input**

100248

**Output**

8

**Explanation**

The digits are **1, 0, 0, 2, 4, 8**.

The largest digit is **8**.



```
#include<stdio.h>
#include<conio.h>
```

```
void main()
{
    long long int n;
    int rem,max=0;

    clrscr();

    printf("enter a number:");
    scanf("%lld", &n);

    while (n>0)
    {
        rem=n%10;

        if (rem>max)
        {
            max=rem;
        }
    }
}
```

```
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    }  
    n=n/10;  
}  
printf("largest digit=%d",max);  
getch();  
}
```

enter a number:472953  
largest digit=9\_



## MEDIUM LEVEL:

### Problem Statement

Given two positive integers **L** and **R**, count how many prime numbers are present in the inclusive range **[L, R]**.

A **prime number** is a natural number greater than **1** that has exactly two distinct positive divisors: **1** and itself.

Your task is to write a program that counts and prints the total number of prime numbers in the given range.

### Input Format

Two integers **L** and **R** separated by a space.

### Output Format

Print a single integer representing the total number of prime numbers in the given range.

### Constraints

$$1 \leq L \leq R \leq 10^5$$

### Example 1

#### Input

1 10

#### Output

4

#### Explanation

The prime numbers between **1** and **10** are: 2, 3, 5, 7

Hence, the total count is **4**.

### Example 2

#### Input

11 20

#### Output

4

#### Explanation

The prime numbers between **11** and **20** are: 11, 13, 17, 19

Hence, the total count is **4**.



```
#include<stdio.h>  
#include<conio.h>
```

```
void main()  
{  
    int l,r,i,j,count=0,prime;  
  
    clrscr();  
  
    printf("enter l and r:");  
    scanf("%d%d", &l,&r);  
  
    for(i=1;i<=r;i++)  
    {  
        if(i<2)  
            continue;  
  
        prime=1;  
  
        for(j=2;j<=i/2;j++)  
        {
```

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```
{  
    if (i%j==0)  
    {  
        prime=0;  
        break;  
    }  
}  
    if (prime==1)  
        count++;  
}  
printf("total prime numbers=%d",count);  
getch();  
}
```

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```
enter l and r:1 10  
total prime numbers=4
```



### Problem Statement

Given a positive integer **N**, find the sum of all its **proper divisors**.

A **proper divisor** of a number is a positive divisor of the number excluding the number itself.

Your task is to write a program that calculates and prints the sum of all proper divisors of **N**.

### Input Format

A single integer **N**.

### Output Format

Print a single integer representing the sum of all proper divisors of **N**.

### Constraints

$$1 \leq N \leq 10^6$$

### Example 1

#### Input

12

#### Output

16

#### Explanation

The proper divisors of **12** are:

1, 2, 3, 4, 6

Their sum is:

$$1 + 2 + 3 + 4 + 6 = 16$$

### Example 2

#### Input

15

#### Output

9

#### Explanation

The proper divisors of **15** are:

1, 3, 5

Their sum is:

$$1 + 3 + 5 = 9$$

```
#include<stdio.h>
#include<conio.h>
```

```
void main()
{
    int n,i,sum=0;

    clrscr();

    printf("enter a number:");
    scanf("%d", &n);

    for(i=1;i<n;i++)
    {
        if(n%i==0)
        {
            sum=sum+i;
        }
    }
}
```

}

printf("sum of proper divisors=%d",sum);

getch();

}

enter a number:12

sum of proper divisors=16\_



**Problem Statement**

Given a positive integer **N**, print all the factors of the number in **ascending order**.

A factor of a number is an integer that divides the number completely without leaving any remainder.

Your task is to write a program that prints all the factors of **N**.

**Input Format**

A single integer **N**.

**Output Format**

Print all the factors of **N** in ascending order, separated by spaces.

**Constraints**

$$1 \leq N \leq 10^6$$

**Example 1****Input**

12

**Output**

1 2 3 4 6 12

**Explanation**

The numbers **1, 2, 3, 4, 6, and 12** divide **12** exactly without leaving any remainder.

**Example 2****Input**

25

**Output**

1 5 25

**Explanation**

The numbers **1, 5, and 25** are the only factors of **25**.

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ADHAYA.CPP

```
#include<stdio.h>
#include<conio.h>
```

```
void main()
{
```

```
    int n,i;
```

```
    clrscr();
```

```
    printf("enter a number:");
    scanf("%d", &n);
```

```
    printf("factors are:");
```

```
    for(i=1; i<=n; i++)
    {
```

```
        if (n%i==0)
```

```
        {
```

```
            printf("%d ", i);
```

```
        }
```

```
    }
```

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ADHAYA.CPP

```
[■]  
}
```

```
    getch();  
}
```

enter a number:12

factors are:1 2 3 4 6 12 \_